Hardware Realization of an 8-Point FFT Using Streaming Pipeline Architecture

Rajesh Reddy
EE24BTECH11043

Department of Electrical Engineering
IIT Hyderabad

1 Introduction

Fast Fourier Transform (FFT) is a fundamental algorithm in digital signal processing for efficient spectral analysis. This work focuses on implementing an 8-point Radix-2 FFT in hardware using a streaming pipeline approach.

The design emphasizes real-time processing capability, reduced memory usage, and efficient fixed-point arithmetic.

2 FFT Formulation

The N-point DFT is defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

For $N = 8$, the FFT computation is decomposed into three stages using Radix-2 decomposition.

The twiddle factors used are:

$$W_8^k = e^{-j\frac{2\pi}{8}k}$$

Key values:

$$W_0 = 1, \quad W_2 = -j, \quad W_1 = 0.707 - j0.707, \quad W_3 = -0.707 + j0.707$$

3 Design Methodology

3.1 Streaming Input Model

- One complex input sample per clock cycle
- Total 8 clock cycles to load complete data
- Continuous processing enabled through pipelining

3.2 Fixed-Point Representation

- Input data: 8-bit signed representation
- Fractional scaling used for precision
- Bit growth handled progressively across stages

4 Architecture Description

4.1 Stage 1: Initial Butterfly

- Temporary storage of first half inputs
- Parallel computation of sum and difference
- Output forwarded to next stage without stall

4.2 Stage 2: Intermediate Processing

- Combines outputs from Stage 1
- Implements special-case twiddle $(-j)$ efficiently
- Produces partially transformed outputs

4.3 Stage 3: Final Transformation

- Performs complex multiplication with twiddle factors
- Uses shift-based scaling for fixed-point multiplication
- Outputs FFT coefficients in pairs

5 Pipeline Characteristics

5.1 Latency Analysis

The architecture introduces initial delay due to:

- Input buffering
- Stage-wise processing delay

5.2 Throughput Analysis

Once steady state is reached:

- Continuous output generation is achieved
- Effective throughput equals one sample per clock

5.3 Critical Path

The longest combinational delay occurs in:

- Complex multipliers in the final stage

6 Simulation and Verification

6.1 Functional Verification

- Testbench applies sinusoidal inputs
- Outputs captured and reordered
- Verified against numerical FFT computation

6.2 Result Comparison

- Output spectrum matches expected frequency bins
- Minor deviations due to quantization effects

6.3 Comparison with FFT IP Core

- Functional equivalence observed
- IP core shows improved timing performance

6.4 Verification Snapshots

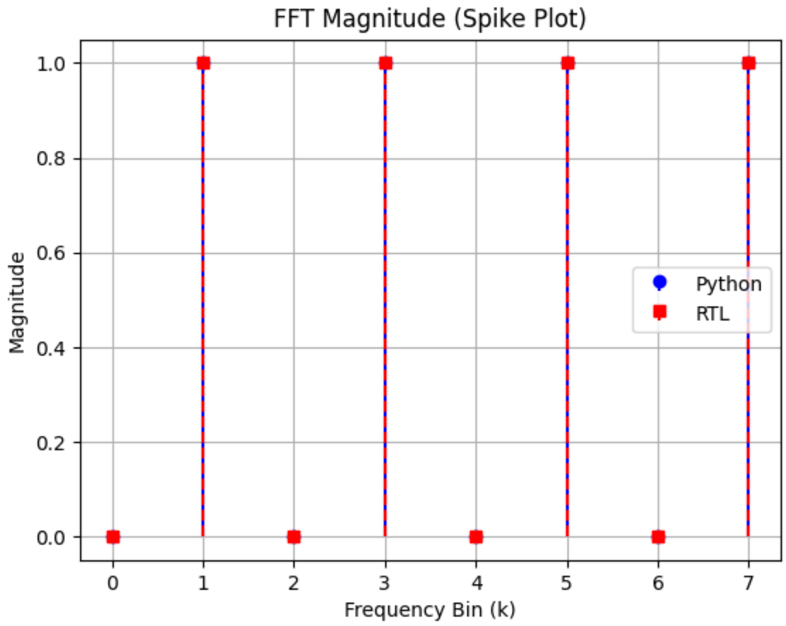

Figure 6.1: Magnitude

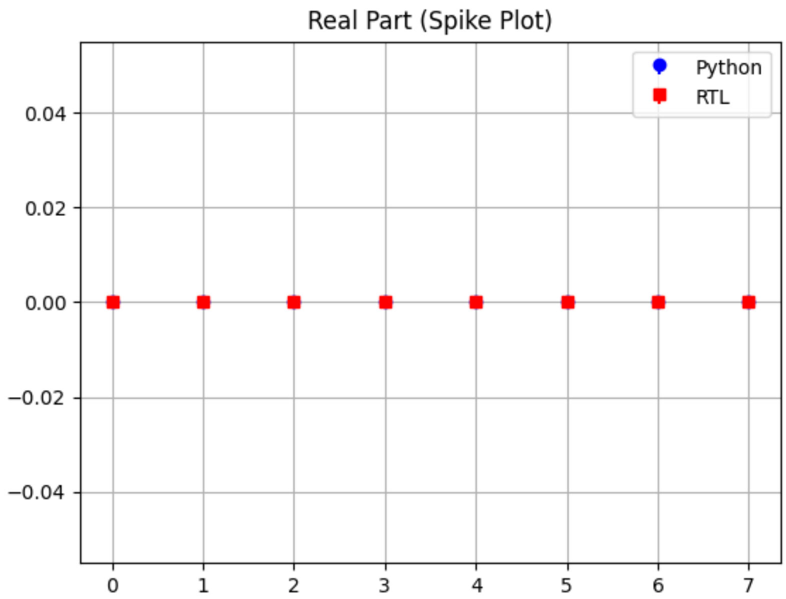

Figure 6.2: Real Part

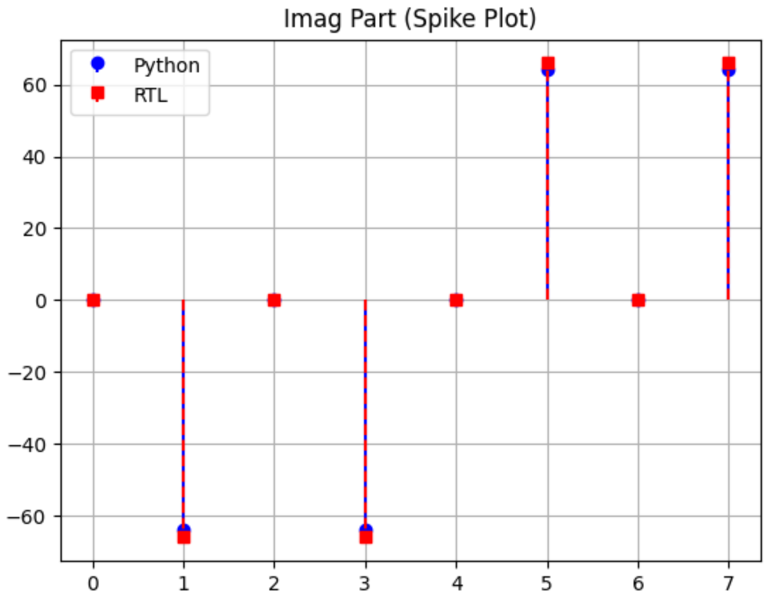

Figure 6.3: Imag Part

7 Discussion

The streaming FFT architecture eliminates the need for large memory buffers and supports real-time data processing. However, performance is limited by multiplier delay in the final stage.

8 Conclusion

An 8-point FFT processor was successfully implemented using a pipelined streaming architecture.

Key outcomes:

- Efficient pipeline utilization
- Accurate FFT computation
- Trade-off between speed and hardware complexity

Future improvements may include higher radix architectures and optimized multiplier design.